



Every order heard. Every loop closed.

The voice recorder for cardiac arrests. It hears the whole resuscitation team, logs every drug, shock and rhythm, runs the ACLS clock and speaks up when an order was never confirmed back.

THE PROBLEM

A code is run entirely by voice, and **orders get lost.**

26%

of 387 spoken orders in 89 filmed resuscitations were closed-loop (repeated back).

El-Shafy et al., J Surg Educ 2018

3.6×

faster completion of orders that **were** closed-loop.

Same study, HR 3.6 (95% CI 2.5–5.3)

≈ 0.27

agreement between hospital records and observers on epinephrine timing.

J Clin Med 2023, cardiac ICU arrests

Hands are on the chest, the airway and the syringes. Someone must stop patient care to write. Tap-based recorder apps still need free hands, and **nothing notices an order that nobody confirmed.**

Every order becomes a loop that must close.

1 • ORDERED

“Give one milligram of epinephrine.”

2 • READ BACK

“Epi one milligram, drawing up.”

3 • GIVEN

“Epi's in.”

SILENCE FOR 10 S → AMBER • 15 S → SPOKEN

Nobody confirmed the order.

DIFFERENT VALUE READ BACK → RED

Ordered 300 mg · read back 150 mg

“Epinephrine one milligram ordered. Not acknowledged.”

“Check dose. Ordered amiodarone three hundred milligrams, read back one hundred fifty milligrams.”

It also runs the ACLS clock out loud: 2-minute cycles, the epinephrine window, shocks and a shockable-rhythm check. It answers **“CodeLoop, last epi kab diya tha?”** from its own record.

THE CRASH-CART SCREEN • LIVE REPLAY THROUGH ASSEMBLYAI

REPLAY

Ears live

Voice ready

1.2s lag

00:59

CODE TIME

Mute 2 min

Speaks: timers + loops

Record

End code

CPR CYCLE

01:21

TO RHYTHM CHECK

CYCLES 0

EPINEPHRINE

Next window in 02:51

last dose

3 min

5 min

Last dose 00:08 ago

RHYTHM • SHOCKS • DRUGS

RHYTHM

V-fib

shockable · 00:43 ago

SHOCKS

1

last 200 J · 00:40 ago

Epinephrine 1 mg

00:08 ago

OPEN ORDERS

1

Amiodarone 300 mg

00:03 open

ORDERED

"Amiodarone 300 milligrams" — C

READ BACK DIFFERS

"Amio 150 pushing" — C

GIVEN

—

Ordered 300 mg · read back 150 mg

Confirm 300 mg

Confirm 150 mg

COMPLETED

00:54

Epinephrine 1 mg

confirm value

went unacknowledged

closed loop

00:22

Shock 200 J

no order heard

TAP TO LOG

Pause CPR

Shock 200 J

Epi 1 mg given

Amio 300 mg given

Rhythm VF

Rhythm PEA

Rhythm asystole

HEARD IN THE ROOM

Arjun: compressor

B

Resume compressions. 00:24

CPR RESUME

meds

Give 1 milligram of epinephrine. 00:31

ORDER epinephrine 1 mg

B

Is IV access in yet? 00:33

meds

Line is in the left arm, 20 gauge. 00:36

C

Sorry, epi, one palayshi release of mercury ki ririki ki postmenentum fil of fil fil philadelphia chinu kinne barmanian. Epi one milligram is in. 00:54

READ BACK epinephrine 1 mg ?

GIVEN epinephrine 1 mg ?

low confidence

C

Amiodarone 300 milligrams. 01:00

ORDER amiodarone 300 mg

C

Amio 150 pushing. 01:01

READ BACK amiodarone 150 mg

CodeLoop

Check dose. Ordered amiodarone three hundred milligrams, read back one hundred fifty milligrams.

Waiting for a gap in the room...

Ask CodeLoop: last epi? what's due?

Ask

Real screenshot: the amiodarone read-back conflict caught at 00:59, with CodeLoop's spoken line on the right

4

Two AssemblyAI connections, each doing a job nothing else could.

CAPABILITY	WITHOUT IT	MEASURED
Universal-3.5 Pro streaming + tight turn detection	The record lags the code	event latency ≈ 1.2 s
Medical Mode + ACLS keyterms + context prompt	Drug names and doses misheard	96% vs 78% critical entities; the dose conflict was missed without them
Word-level speaker labels	Order and read-back look like one person	Turns split at word-level speaker changes
Word confidence	A misheard dose silently becomes fact	Below 0.6 → UNCONFIRMED, listed for sign-off
Hindi-English code-switching	Indian teams' real talk lost	Cross-script phonetic layer maps Devanagari back to English
Voice Agent API (semantic barge-in)	A bot talks over the team leader	~0.8 s to speak; stops mid-word when interrupted

Models understand speech. **Deterministic code decides.**

AssemblyAI · ears

U3.5 Pro Streaming

Room audio from a crash-cart tablet.
Speaker labels, Medical Mode, ACLS
keyterms, far-field Voice Focus, English
+ Hindi.

CodeLoop core

Grammar → resolver → ACLS engine

Every event quotes the exact words it came from.
Loops, timers, flags and every spoken number come
from unit-tested, replayable code. **The model never
picks a dose.**

AssemblyAI · voice

Voice Agent API

Speaks fixed lines only, waits for a gap
in the room, stops when a clinician
talks over it. Never hears the room's
chatter.

Every utterance, event, loop, flag and spoken line → a **SHA-256 hash-chained audit log**. The Code Record is rebuilt from it, so the screen and the record cannot disagree.

Built so that being wrong is **visible**, not silent.

- **Low ASR confidence** on a dose → UNCONFIRMED, never a fact
- **Read-back differs** → red card, one-tap confirm, spoken check
- **A misheard “ROSC”** only pauses timers; a human ends the code
- **CodeLoop's own voice** is recognised and never logged as the team
- **Never recommends treatment:** it reminds timing, never what to give
- **Answers come from the record:** a managed LLM chose the wrong fact in testing, so answers are composed deterministically
- **Tamper-evident record** with integrity check and head hash
- Screen-only / timers-only / mute: the team decides how chatty it is

Held-out mock codes, committed before they were measured.

LIVE ASSEMBLYAI	PRECISION	RECALL	LOOPS
Held-out set 1, first run	0.97	0.71	5 / 8
Held-out set 2, first run	0.82	0.56	7 / 9
Both, after general fixes	0.96	0.88	16 / 17
VF demo, production config	0.97	0.91	3 / 3

- Each fresh set found real bugs, including “**one hundred and fifty joules**” → **two shocks**. Fixed with general rules plus a regression test.
- When unsure, CodeLoop **misses** rather than invents: precision stays high.
- Remaining misses are mostly recognition errors (“two hundred” heard as “100”), exactly what the read-back loop exists to catch.
- Limitation: synthetic TTS voices so far. Real-voice testing in a sim centre is milestone one.

Start where every hospital already practises: **mock codes.**

1 • Simulation

Sim centres and resuscitation committees: debrief timelines and closed-loop scores for every mock code. No patient data, no clinical reliance.

2 • Documentation

Live codes, silent or timers-only. The recorder nurse reviews and signs; FHIR and GWTG-Resuscitation registry export.

3 • Real-time assist

Spoken loop nudges during live codes, after clinical validation and a SaMD regulatory pathway.

≈ **\$2.66** of AssemblyAI usage per 30-minute code (streaming + Medical Mode + diarization + Voice Focus + Voice Agent), at list prices.

Per-site subscription for simulation; per-bed for clinical documentation. Expands to trauma, pediatric and neonatal resuscitation, and EMS.



Codes are loud, fast and hands-full.
CodeLoop is the recorder that **never looks away.**

Every order heard. **Every loop closed.**

`github.com/TusharTechs/codeloop · live demo: see submission`